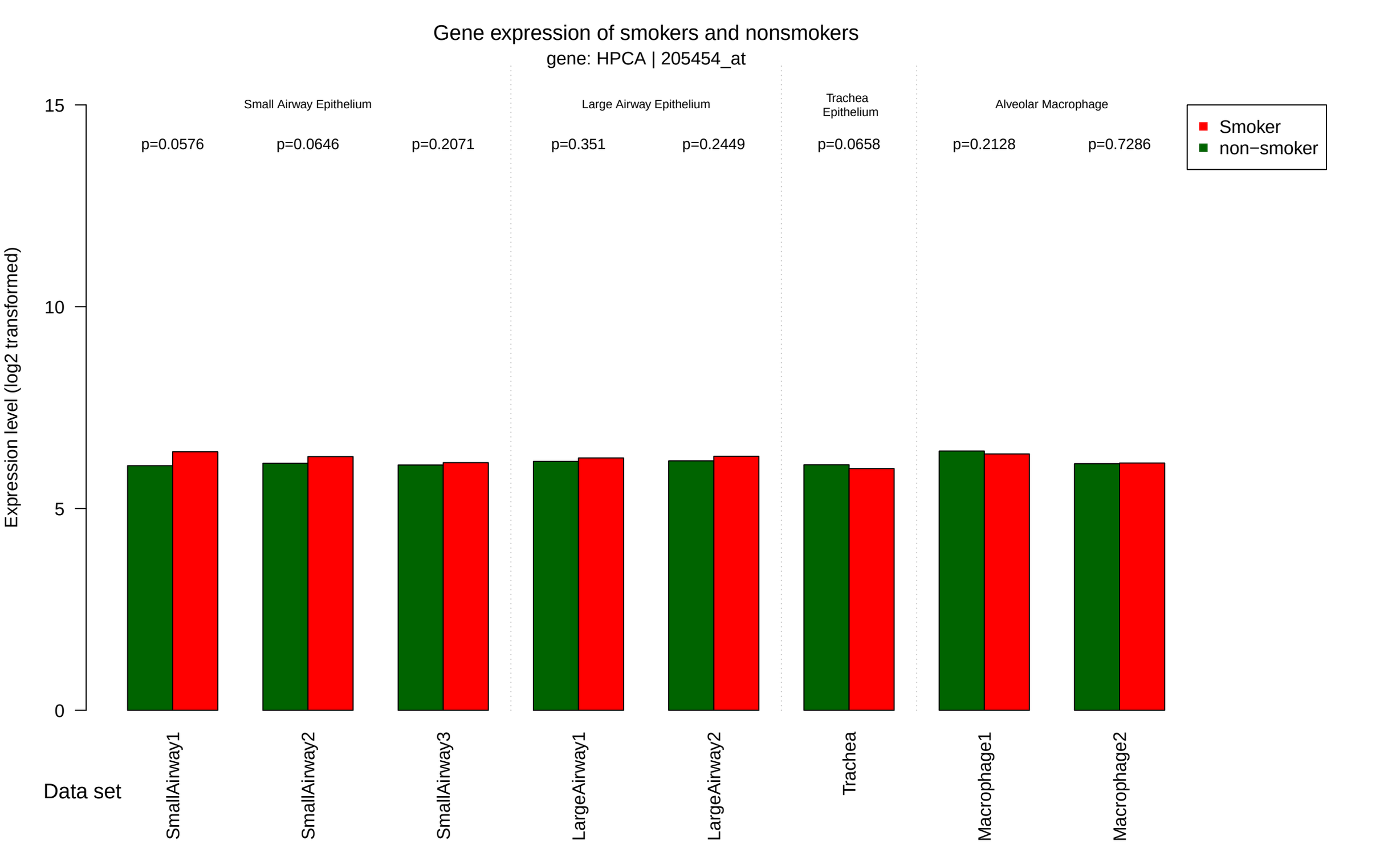
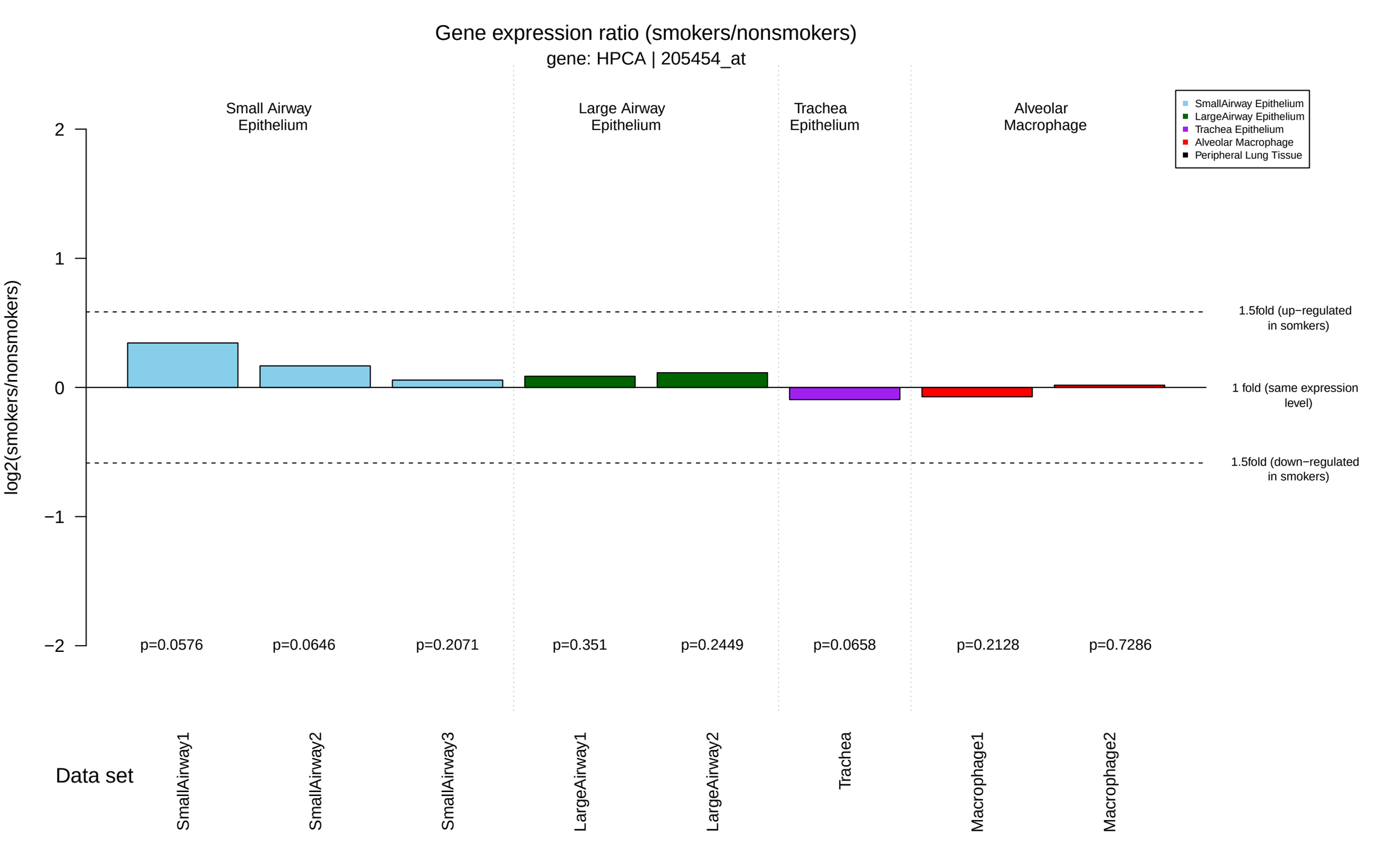
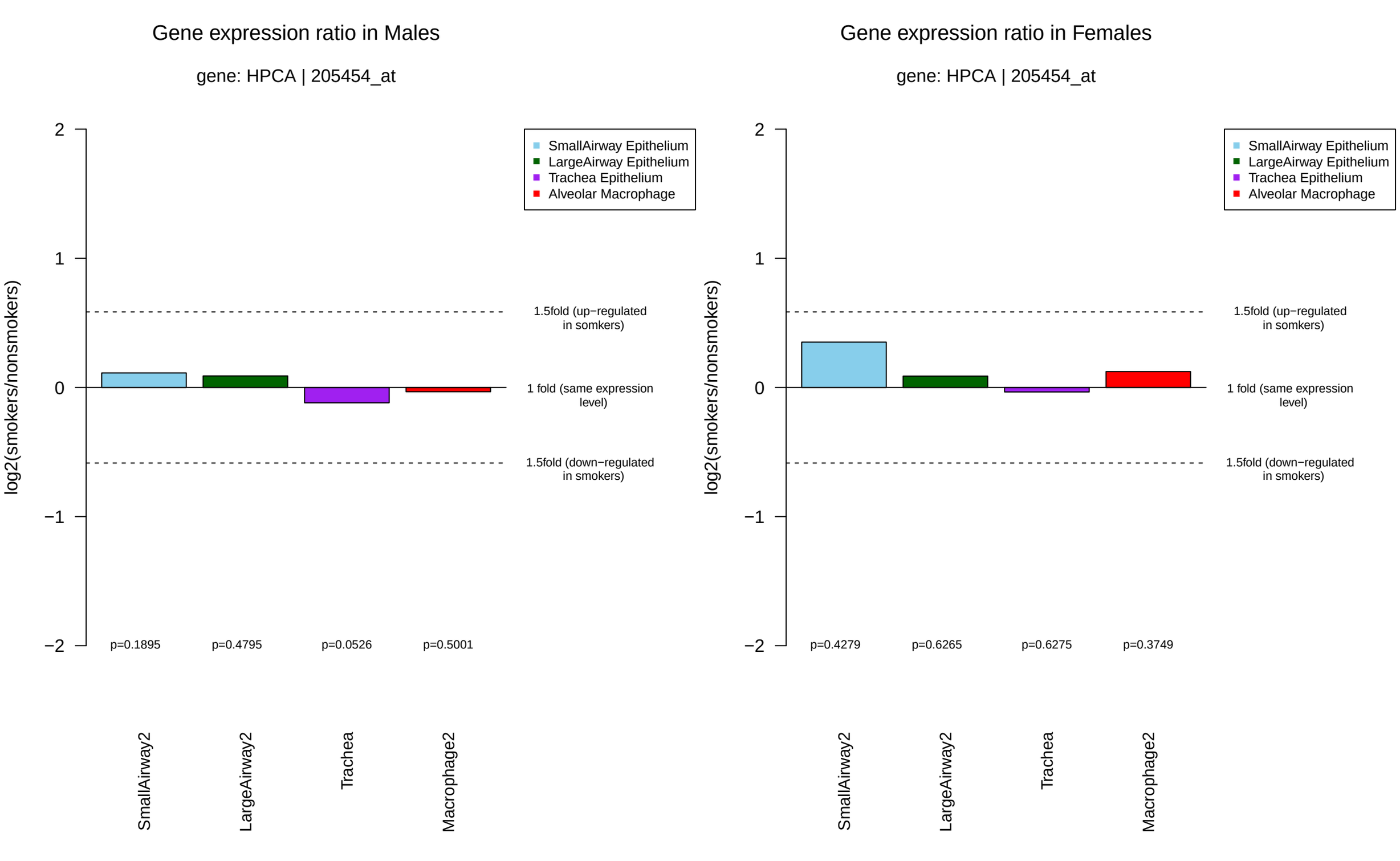
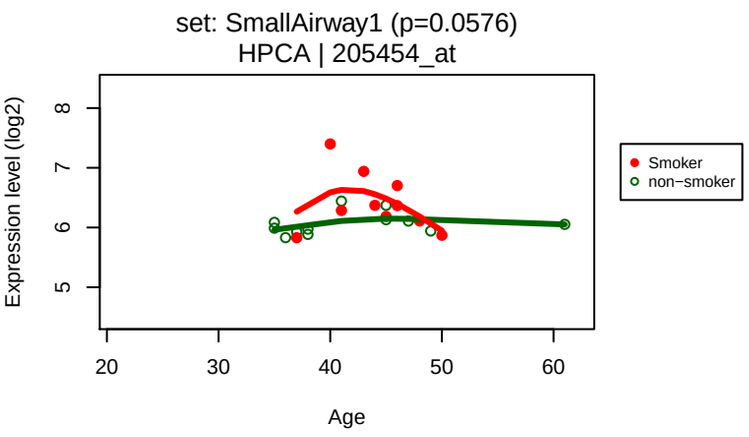
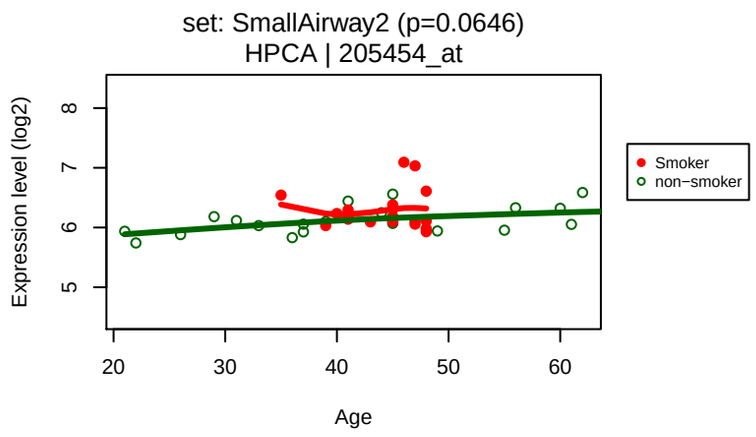
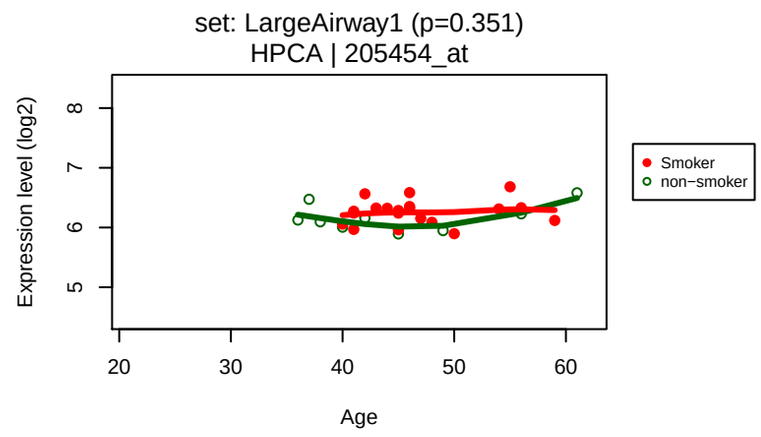

Gene expression of smokers and nonsmokers
gene: HPCA | 205454_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0576
p=0.0646
p=0.2071
p=0.351
p=0.2449
p=0.0658
p=0.2128
p=0.7286
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

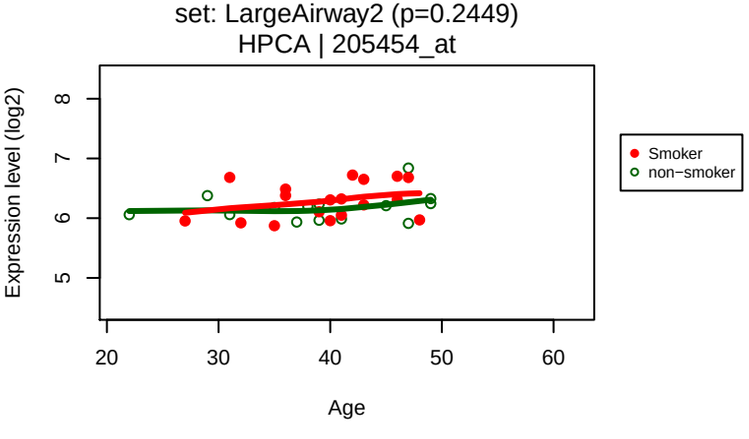

Gene expression ratio (smokers/nonsmokers)
gene: HPCA | 205454_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0576
p=0.0646
p=0.2071
p=0.351
p=0.2449
p=0.0658
p=0.2128
p=0.7286
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

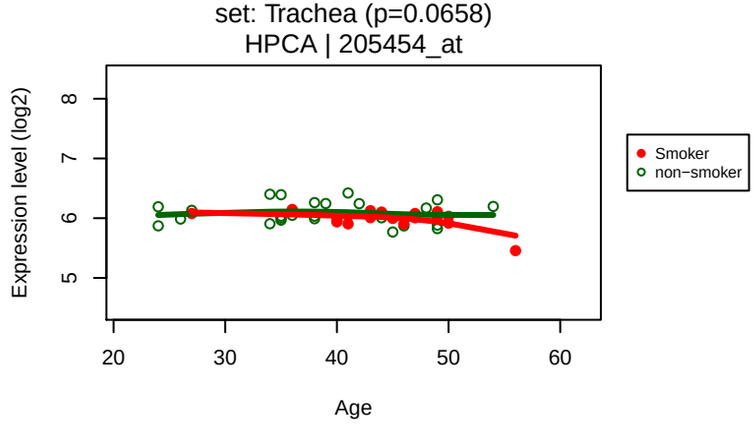

Gene expression ratio in Males
gene: HPCA | 205454_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1895
p=0.4795
p=0.0526
p=0.5001
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HPCA | 205454_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4279
p=0.6265
p=0.6275
p=0.3749
SmallAirway2
LargeAirway2
Trachea
Macrophage2

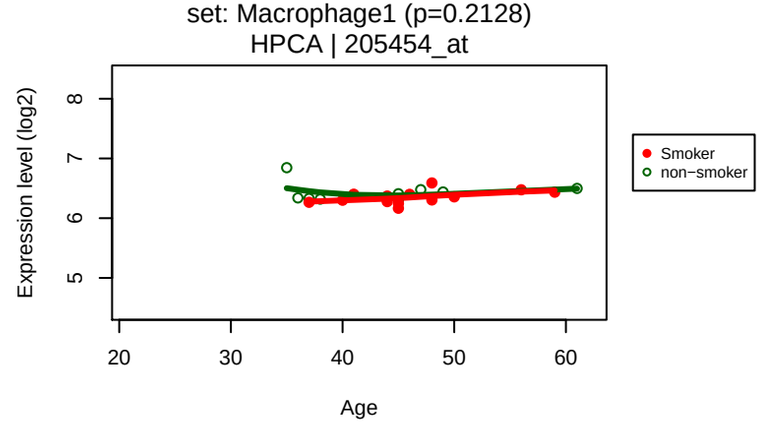

set: SmallAirway1 (p=0.0576)
HPCA | 205454_at
Expression level (log2)
Age
Smoker
non−smoker

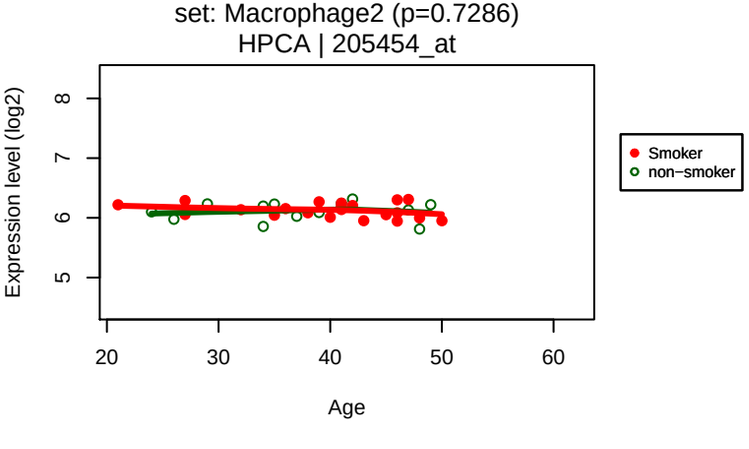

set: SmallAirway2 (p=0.0646)
HPCA | 205454_at
Expression level (log2)
Age
Smoker
non−smoker

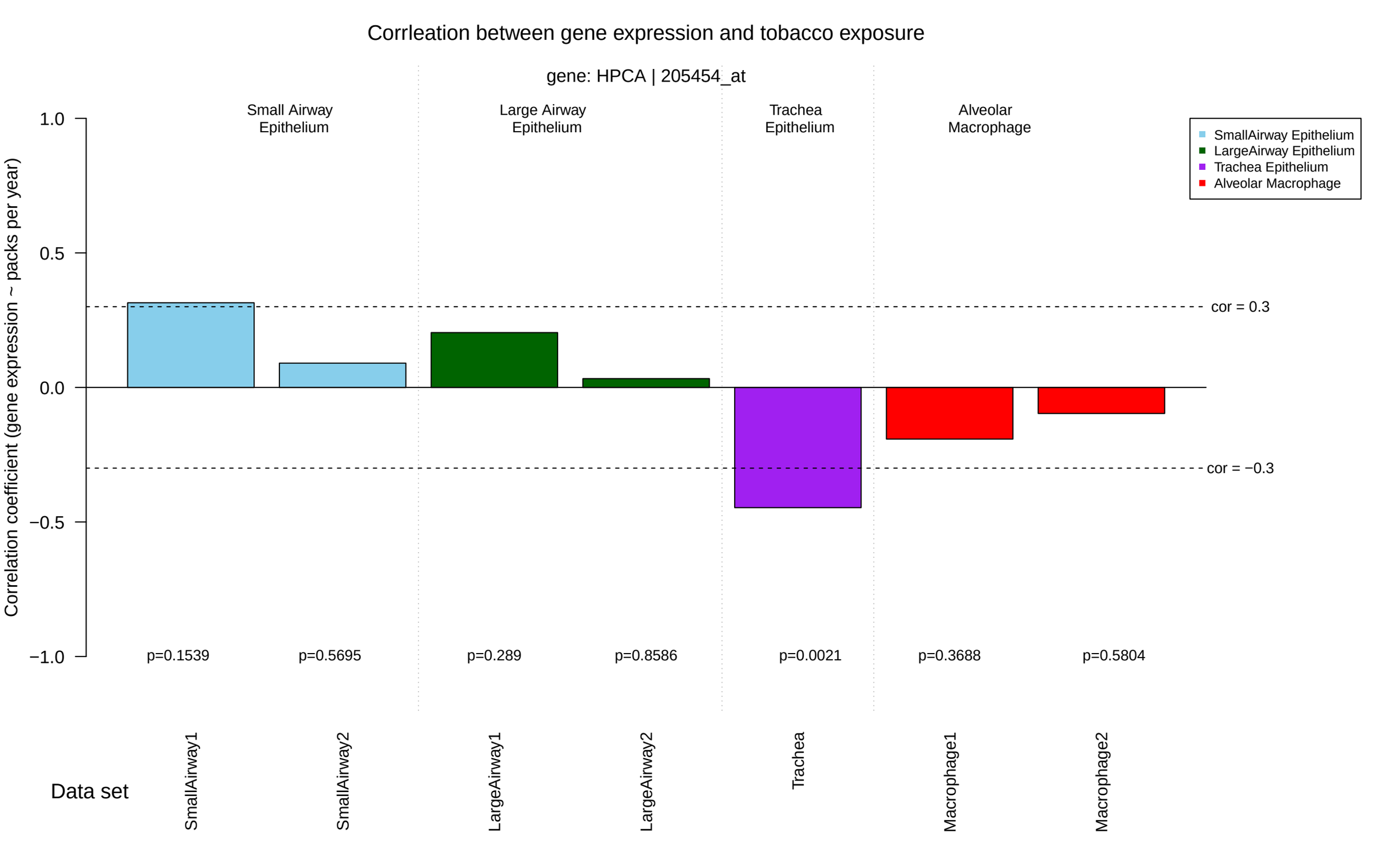

set: LargeAirway1 (p=0.351)
HPCA | 205454_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2449)
HPCA | 205454_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0658)
HPCA | 205454_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2128)
HPCA | 205454_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7286)
HPCA | 205454_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HPCA | 205454_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1539
p=0.5695
p=0.289
p=0.8586
p=0.0021
p=0.3688
p=0.5804
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2